Assignment 1 – Python Programming

Course Code: BES00007

Question 1. *Discuss the use of break and continue statements in Python loops.*

Answer:

break statement: It is used to terminate the loop immediately, regardless of the loop's original condition. Once break is executed, control jumps to the statement right after the loop.

Example:

```
for i in range(1, 10):
    if i == 5:
        break
    print(i)
```

Output: 1 2 3 4

continue statement: It skips the remaining code inside the loop for the current iteration only, and moves on to the next iteration. The loop does not terminate.

Example:

```
for i in range(1, 6):
    if i == 3:
        continue
    print(i)
```

Output: 1 2 4 5

Comparison: break vs continue

Aspect	break	continue
Action	Terminates the loop immediately	Skips only the current iteration
Loop control	Exits the loop entirely; no further iterations occur	Loop continues with the next iteration
Code after it (in the block)	Not executed for the current or any later iteration	Not executed for the current iteration only
Typical use	Stop searching once a condition/target is met	Skip invalid or unwanted values and keep processing
Effect on output	1 2 3 4 (stops at i = 5)	1 2 4 5 (skips i = 3)

Question 2. *Describe the string methods with examples and explain their functions.*

Answer:

1. upper() and lower() - Convert a string to uppercase or lowercase.

```
s = "Hello World"
print(s.upper())   # HELLO WORLD
print(s.lower())   # hello world
```

2. strip()- Removes leading and trailing whitespace from a string.

```
s = "   Python   "
print(s.strip())   # "Python"
```

3. split() - Splits a string into a list of substrings based on a delimiter.

```
s = "apple,banana,cherry"
print(s.split(','))   # ['apple', 'banana', 'cherry']
```

4. replace(old, new) - Replaces all occurrences of a substring with another substring.

```
s = "I like cats"
print(s.replace("cats", "dogs"))   # I like dogs
```

5. find(sub) - Returns the lowest index of the substring if found, else -1.

```
s = "Python programming"
print(s.find("prog"))   # 7
```

6. len() - Although not a string method, it returns the number of characters in a string.

```
s = "Python"
print(len(s))   # 6
```

7. capitalize()- Converts the first character of a string to uppercase.

```
s = "python programming"
print(s.capitalize())   # Python programming
```

8. title() - Converts the first character of each word to uppercase.

```
s = "python programming language"
print(s.title())   # Python Programming Language
```

8. count() - Returns the number of times a substring occurs in a string.

```
s = "banana"
print(s.count("a"))   # 3
```